Team teaching material

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1 Welcome

This site is built with Quarto as a **book** project. Use the table of contents to open the Linux installation guide.

Download the compiled book as a single PDF: team-teaching-material.pdf.

For other materials in this repository (SNF, Git, package development, etc.), see the project `README.md` in the repository root.

2 Install core software on Ubuntu

- `dpkg --print-architecture`: retrieve architecture, for package installation[1].
- `Ctrl-H` in **Gnome**, aka File Management, to list hidden files

2.1 Ubuntu general guidelines

2.1.1 Recommended system libraries to install

```
# Install system libs
sudo apt-get update
sudo apt-get install -y \
    apt-transport-https \
    build-essential \
    ca-certificates \
    cloud-guest-utils \
    curl \
    default-jre \
    f2c \
    gdebi-core \
    gnupg \
    libbz2-dev \
    libcurl4-openssl-dev \
```

[1] Returns `amd64` in my case

```
    libffi-dev \
    libfontconfig1-dev \
    libfribidi-dev \
    libgsl0-dev \
    libgtextutils-dev \
    libharfbuzz-dev \
    libicu-dev \
    liblzma-dev \
    libncurses5-dev \
    libncursesw5-dev \
    libpcre3 \
    libpcre3-dev \
    libperl-dev \
    libreadline-dev \
    libssl-dev \
    libtiff-dev \
    libxml2-dev \
    locales \
    ncftp \
    pandoc \
    parallel \
    pkg-config \
    readline-doc \
    sudo \
    tcl \
    tk \
    tk-dev \
    tk-table \
    tmux \
    tzdata \
    unzip \
    vim \
    && sleep 1
```

2.1.2 Ubuntu file, and package structure

2.1.2.1 System-wide location (for core programs)

System-wide affects all users, are installed using `sudo`, on the root of the system.

Location	Purpose
`/usr/bin`	System executables (APT)
`/etc`	System-wide configuration (example: define repositories for `apt`)
`/opt`	Vendor (external) apps
`/usr/local/bin`	Admin-installed software
`/var`	Logs, caches, databases

Table 2.1: Main system-wide software locations

- `/opt` for third-party application bundles, **self-contained** while `/usr/local` is for third-party **binaries**. Besides, `/opt` is not in the path by default.

2.1.2.2 User-specific locations

User does not require root

Location	Purpose
`~/Applications`	AppImages
`~/.local/bin`	User executables
`~/scripts/tools`	Custom bash executables

Location	Purpose
`~/.local/share`	User app data, notably `~/.local/share/applications` storing desktop icons
`~/.config`	User configuration

Table 2.2: Main user-based software locations, to be compared with Table 2.1

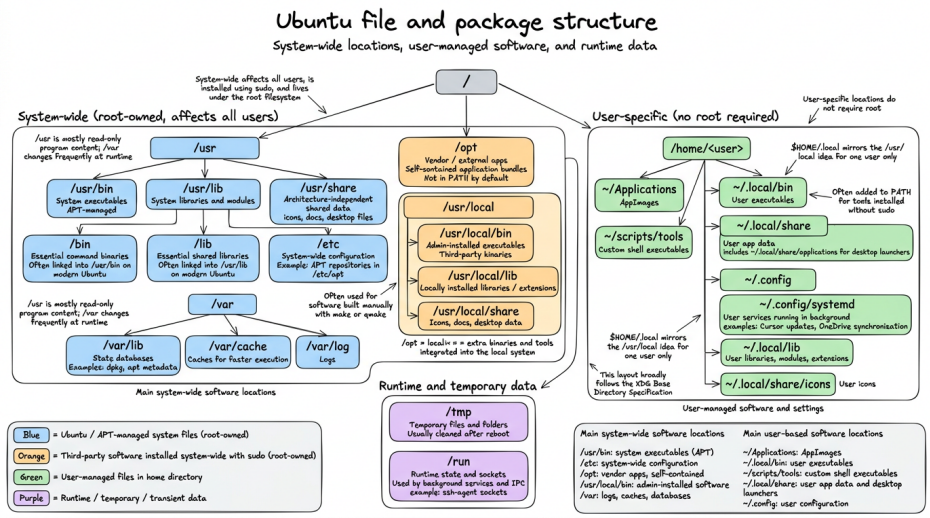


Figure 2.1: Detailed structure of Ubuntu package layout

Specific folders

- `usr` folder stores *read-only* executables, while `/var` is regularly written upon changes:
  - `/var/lib`: apt and dpkg databases, for instance, respectively stored under `dpkg`, and `/apt`
  - `/var/cache`: metadata caches for faster execution
- `/usr/local/`, or `$HOME/.local` stores executables that require custom compilation with `make`, or `qmake`, rendered from manual archives:
  - `bin`, `sbin` for executables
  - `share` for icons (under subfolder `icons`) and desktop configuration, document ion under `/doc`
  - `lib` for extensions of a given executable (such as packages in R, or modules in Python)
  - This folder hierarchy follows `XGD`Base Directory Npecification
- `/tmp` for temporary folders, cleaned after `reboot`
- `/run` for sockets, namely used for controlling all the protocols running in the background for connecting to a server, such as the `ssh-agent`
- `~/.config/systemd`: lists all processes that keep on running in the background, such as Cursor updates, or One drive synchronisations.

2.1.3 Recommended Updating policy of Ubuntu

1. Create file `update_os.sh` with:

```
#!/bin/bash
rm -rf /var/lib/dpkg/lock-frontend
rm -rf /var/lib/dpkg/lock
apt-get update
apt-get upgrade -y
apt-get dist-upgrade -y
apt-get autoremove -y
apt-get autoclean -y
```

2. Run having `sudo`rights

i. `chmod +x update_os.sh` to provide executive rights
ii. `sudo ./update_os.sh` to run the update file with **admin** rights

3. Reboot the system with `reboot`

Ubuntu partial updates

Partial updates breaking key driver components occur more often with Ubuntu compared to Microsoft, since Linux tends to provide lots of separate, modular packages while Windows ships large and standalone bundles -> **Consequence: more issues with updates, especially if the network connection dropped, and the newest versions of the packages were not fully installed.**

General recommendations:

```
# to be run daily
sudo apt update
sudo apt upgrade

# to be run monthly
sudo apt full-upgrade

## clean deprecated dependencies
sudo apt autoremove
sudo apt autoclean
```

Deal driver issues

```
## Check for Broken Packages
sudo dpkg --configure -a
## Install missing dependencies
sudo apt -f install

## List all Hardware Driver Errors
dmesg -l err,crit,alert,emerg
```

2.1.4 Monitor processes running in the background, and on startup

- `systemctl list-unit-files --type=service --state=running --user` to list all processes running in the background, even after rebooting, restrained to the user-level.
  - Alternatively, processes at the system-level are stored under `/lib/systemd/system/`

- `systemctl status <service>` retrieves current status

2.1.5 Network protocols, and connexion troubleshoots

- Two IP protocols for Wi-FI: `IPv4` versus `IPv6`.

> The two Internet protocols, namely IPv4 vs IPv6
>
> IPv4 is older, but still prevailing in many websites, such as Github, or OneDrive:
>
> – Uses 32-bit addresses (e.g. `192.168.1.10`)
> – Used by **most websites**, CDNs, VPNs, corporate services
> – But the 32-limitation hinders the number of avalaible IP addresses.
>
> IPv6 is newer, but not universally supported:
>
> – Uses 128-bit addresses (e.g. `2a01:cb14:86a7::1`)
> – Supported by modern ISPs and operating systems
> – Some networks provide **IPv6-only**, with no IPv4 fallback. Espaically the case with Microsoft services, governmental or academic websites.

- You can test if this is the issue to connect to some websites with `curl <webdomain>`, such as `curl https://onedrive.live.com`, returning *Cannot connect*
- Recent modern routers return only IPv6 -> while Windows is able to able to auto-fall back to IPv4, this is not the case of IPv6.
  – *Temporary fix*: by running the following instructions, you solve this issue by automatically configuring a IPv4 upon failure. *Recommended only for short-term debugging.*

```
install dhclient commands
sudo dhclient -4 wlp2s0f0 <Wifi-name> ## in my case, wlp2s0f0
```

- *Permanent fix*[2]:

```
nmcli connection modify "Routeur/Livebox-FC77" ipv4.method auto
nmcli connection modify "Routeur/Livebox-FC77" ipv6.method auto ## connect automatically to
## re-start the connection, applying the new configuration parameters
nmcli connection down "Routeur/Livebox-FC77"
nmcli connection up "Routeur/Livebox-FC77"
```

2.1.5.1 Eduroam connexion

0. **Ensure to delete beforehand any pre-existing installation:** `rm -rf $HOME/.config/cat_installer`.
1. Go to https://cat.eduroam.org/, the website usually provides with the version fitting your OS distribution.
2. On Linux, download the Python script, choosing AMU university.
3. Run Python script with `python 3 eduroam-linux-AUA.py`
4. As `userid`, provide your academic mail address, and as password, your `ENT password`.

2.1.6 Pimp your shell: switch from boring Bash to fancy `Zsh`

1. Installation script:

```
sudo apt update
sudo apt install zsh

## change the default shell shebang
chsh -s $(which zsh) ## modify /etc/passwd repository
```

[2]Alternatively, you can rely on the GNOME Interface, `Wifi` section

```
reboot # required log out for effective changes

## check if zsh is used by default (also the theme is modifed)
getent passwd "$USER" -> return "/home/bastien:/usr/bin/zsh"

## Install advanced customisation of the shell with Install Oh My Zsh addon
sh -c "$(curl -fsSL https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/in
```

You should keep the original ~/.bashrc as a fallback for Bash.

2. Install the modern alternative to `ls`, namely `eza`

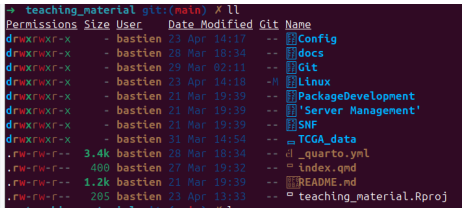
```
# install the gpg command
sudo apt update
sudo apt install -y gpg

# install eza
wget -qO- https://raw.githubusercontent.com/eza-community/eza/main/deb.asc | sudo gpg
echo "deb [signed-by=/etc/apt/keyrings/gierens.gpg] http://deb.gierens.de stable main
sudo chmod 644 /etc/apt/keyrings/gierens.gpg /etc/apt/sources.list.d/gierens.list
sudo apt update
sudo apt install -y eza
```

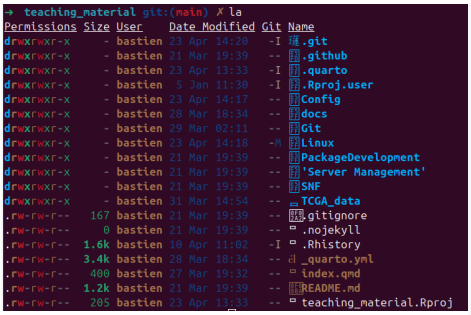
3. Change the behaviour of the shell:

- **System-wide**: all exe installed under `/usr` folder can be called directly from the `$PATH`. For third-party apps stored under `/opt`, you must **add a symlink** redirecting to your local folder: `sudo ln -s /opt/<appName> /usr/local/bin/<appName>`.
- **User-level**: customise `.bashrc` (and `.zshrc` if using the `.zsh`fancy shebang) -> **Only affects the current user**, see my own `.zshrc` configuration file, and below `ls`output Figure 2.2.

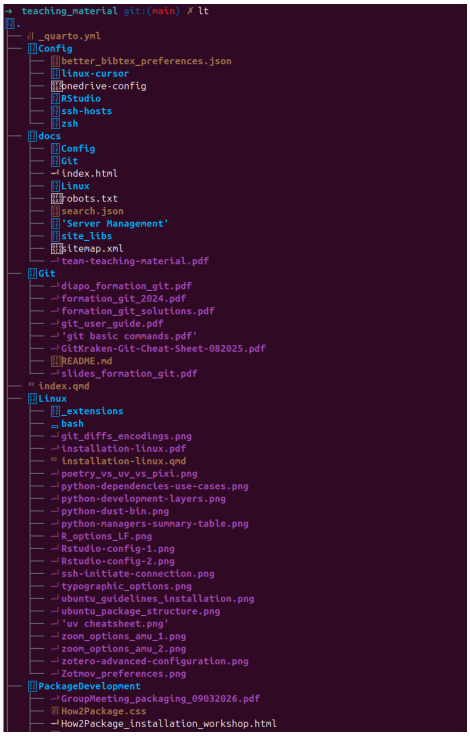
(a) Modified bahaviour of the `ls` output



(b) Modified bahaviour of the `ll` output, displaying addition to file names their sizes, permission rights, and original proprietary



(c) The `la` output displays in addition to the `ll`output the hidden files (traitdionally starting with a `.`)



(d) The `lt` output is combination of both `ls`and `tree`to display the hierarchical structure of a folder

Figure 2.2: Visual representation of my customised `ls`commands to display the content of a folder from a Bash terminal.

Useful resources:

- shell shortcuts

2.2 Synchronise files across devices with OneDrive

2.2.1 OneDrive open-source installation for Ubuntu

1. Install dependencies:

```
sudo apt install build-essential libcurl4-openssl-dev pkg-config git -y
```

2. Add the `OpenSuSE` Build Service repository release key

```
wget -qO - https://download.opensuse.org/repositories/home:/npreining:/debian-ubuntu-

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/obs-onedri

## update cache
sudo apt-get update
```

3. Install `OneDrive`:

```
sudo apt install --no-install-recommends --no-install-suggests onedrive
```

4. Sync OneDrive:

i. Exclude large folders, and directories that are complex and are not recommended for syncing, `nano ~/.config/onedrive/config`. You can see the actual content of my config file under onedrive-custom-config.[3]

ii. Authenticate, and run OneDrive once

[3]Note that **Comments must be placed in separated lines, not inline**.

```
onedrive --sync --dry-run ## optional, list folders that will be synchronsied
onedrive --sync --verbose ## first initialisation, usually time-consuming, one-shot
```

4. Permanent synchronization using `systemtcl` facilities. **Choose either this approach, or the visual approach detailed in Tip 1, as they can't operate simultaneously**.

```
systemctl --user enable onedrive
systemctl --user start onedrive ## in the back end, use onedrive --monitor
```

5. Monitor currently uploaded files:

```
journalctl --user-unit onedrive -f (continous) ## equivalent to systemctl --user status onedrive (sing

onedrive --display-config
onedrive --version
```

6. Other useful OneDrive instructions:

- `onedrive --create-share-link <path/to/file>` for read-only shareable links, and `onedrive --create-share-link <path/to/file> --with-editing-perms` to add writing rights.

> Tip 1: GUIs for OneDrive monitoring
>
> OneDrive system tray program adds a simple icon returning the status of synchronization for OneDrive.
>
> Installation bash:

```
## install the 5th version of qmake
sudo apt install qt5-qmake qtbase5-dev

## create the runnable executive file
git clone https://github.com/DanielBorgesOliveira/onedrive_tray.git
cd onedrive_tray
mkdir build
cd build
/usr/lib/qt5/bin/qmake ../systray.pro ## prepare the configuration, such as Makefil
make ## build the software

## execute it, such that it runs whenever log in
sudo make install ## deploy the software, such as copying files -> here's the sudo
systemctl enable --user onedrive_tray.service
```

Colour the logs output reported on the terminal:

1. Install dependencies: `sudo apt install silversearcher-ag ccze`
2. Add `onedrive_log` bash file into folder `~/scripts/tools`.
3. Add permanently to the `$PATH` by modifying the `~/.bashrc` file: `export PATH="$PATH:$HOME/scripts/tools"`/

2.2.2 Troubleshoots for OneDrive

- **Call `systemctl --user restart onedrive` after modifying the `.config` file to account for changes within it**. For retro-application of configuration changes, use `onedrive --resync --sync --verbose` with parsimony. **The synchronisation halts without warning, when the config file is modified, so you need to restart it**.

- Warning "onedrive application is already running, please check process list" means that several concurrent onedrive are running in the background:
  - List existing processes using onedrive: `ps aux | grep onedrive`
  - Force kill processes that do no not use `systemtcl`: kill -9
  - 90% of the time is that you're simultaneously running UI OneDrive, and in the background `systemtcl` for synchronising processes

i OneDrive, Git and EOL encoding across OS

Git and OneDrive do not combine well for:

- **Different encoding of end of line**: Linux (and Mac) machines end lines with `LF (Line Feed): \n`, while Windows uses `CRLF (Carriage Return + Lien Feed): \r\n`
- Linux tracks file permissions for executables while Windows doesn't (only problematic for shell scripts)
- `.git` and `.Rproj` folders are complex binary repositories often OS dependent.

To avoid these issues:

i. Configure globally the `.gitconfig` configuration file to allow both end of lines indicators, using `git config --global core.autocrlf input` on Linux and Mac machines collaborating with Windows machines[4], and `git config --global core.autocrlf true` on Windows machines[5].

ii. For further robustness, create a `.gitattributes` dotfile per project[6]:

```
## 3 columns: file extension, file type (text, or binary), eol definition for text
*.sh   text eol=lf
*.py   text eol=lf
*.js   text eol=lf
*.ts   text eol=lf
*.java text eol=lf
```

Then, apply the changes with the following Git commands^[this command first identif

```
git add --renormalize .
git status ## display files being impacted by the change of EOL encoding
git commit -m "Normalize line endings"
```

iii. Configure the editor default end of line encoding:

- *VS Code*: `Shift-Ctrl-P`, then type "Open VS Code Settings", then go to "Files:EOL", and set end of line as `\n`
- *RStudio:* `Tools → Global Options → Code → Saving -> Line ending Conversion → (Posix) LF`.

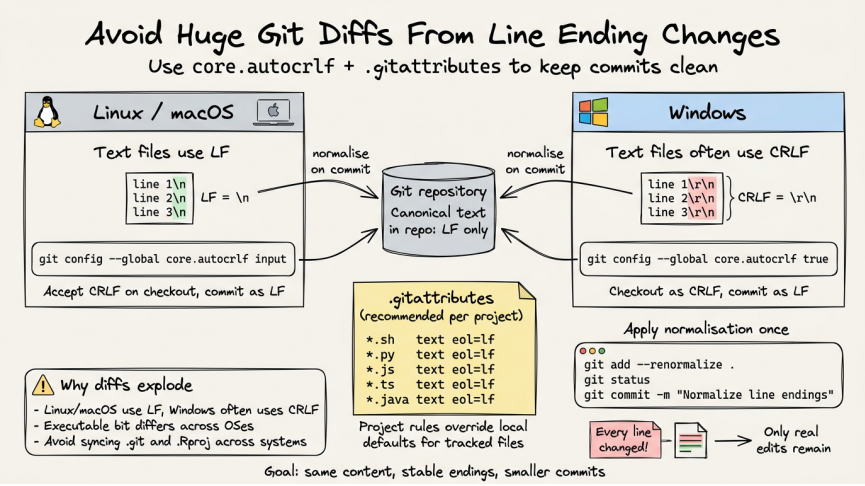


Figure 2.3: Best practices for End of Line homogenisation across devices

> **Graphical Interfaces for OneDrive**
>
> Avoid using graphical interfaces, such as OneDriveGUI, they poorly concur with local installation of `onedrive` on my personal experiment. Instead, rely on the terminal, as it usually provides the user with better customisation, and control on OneDrive configuration.

> **Compatibility with Applications Using Atomic Save Operations**
>
> Many modern editors, such as `LibreOffice` and `Rstudio` use **atomic save strategies** to preserve data integrity when writing files.

[6] Accept as EOL CRLF, but always commit as LF
[6] constrain commits to be encoded and pushed as `LF`
[6] Detailled information is avalaible here

```
file.qmd
↓ edit
file.qmd.tmp ## If OneDrive synchronises in the meantime, it believes that the file
↓ rename
file.qmd


## Architecture causing this problem
RStudio
   │
   │ save
   ▼
Local file (.qmd)
   │
   │ filesystem event
   ▼
onedrive client
   │
   │ sync
   ▼
OneDrive cloud
   │
   │ sync back
   ▼
local overwrite
```

This process is likely to interfere with the regular synchronisation protocol of the `onedrive` client relying on **inotify**, triggering editor warnings, or even temporary removal of files contents.
Three complementary solutions to address these issues, and reduce conflicts between editor saves and sync detection.

1. Do not put active projects inside `OneDrive`, setting apart large

documents storage synchronised with OneDrive, from source code version by Git.
2. Pause sync while coding. **But you need to think about restarting the synchronisation afterwards**.
3. Twist the configuration file, helping with editors using atomic saves:

```
force_session_upload = "true" ## avoid overwriting by older cloud versions, when RStudio is on its v
delay_inotify_processing = "true" ## wait for RStudio to finalise the saving of tmp files before syn
```

2.3 Software to install

**General recommendation*: avoid using `ChatGPT`, as the instructions for installing software are often deprecated and/or not following Ubuntu best guidelines.

Note 1: Install software on Ubuntu, some guidelines

Three recommended methods, depending on Ubuntu's support:

1. If supported by Ubuntu, go for `apt install` for system software[7]:

- *automatic updates* and security patches -> **it's the only method allowing it in a systematic way**.
- easiest method for installation
- signed and verified packages
- recommended for *core system*, long-term software
- If **not directly available on Ubuntu's native package manager, or using in the back-end `snap` commands**, switch

to pre-built tarball installation[8], or add the repository (only if **trusted**) to `apt allowed keys` folder. Example for `Docker`, Section 2.3.6, or custom Zotero installation file.

- **.deb files** are intended for Linux distributions derived from Debian (such as Ubuntu, Linux Mint, etc.), while **.rpm files** are mainly used by distributions derived from Red Hat–based systems (such as Fedora, CentOS, and RHEL) + `openSUSE` distribution:
  - No auto-updates
  - Potential dependency issues
  - **Always install with `sudo apt install`, rather than `sudo dpkg -i`, to better solve dependencies, fix broken installs, and ensures system consistency.**

---

2. `AppImages` are standalone executable files, bundling the application and most of its dependencies:

- Recommended for *desktop applications* to be installed only on the user account -> does not require `sudo` rights in most cases
- Portable
- `AppImageLauncher` can help in organising all AppImages within the same folder + integrate them into the system menus:
  - Relevant if using multiple AppImages
  - Avoid `Snap`

---

3. Pre-built tarball (`.tar(gz|xz)`) archives:

- Stronger customisation

- Works without distribution packages
- Requires manual handling/no automatic updates
- **Only recommended for developers, or if nothing else exists**.

- And now, the least recommended installations:
  - `snap` stores pre-compiled versions of tools, but are less recommended compared to standard installation with `sudo apt install`
  - `FlatPak+Flathub`

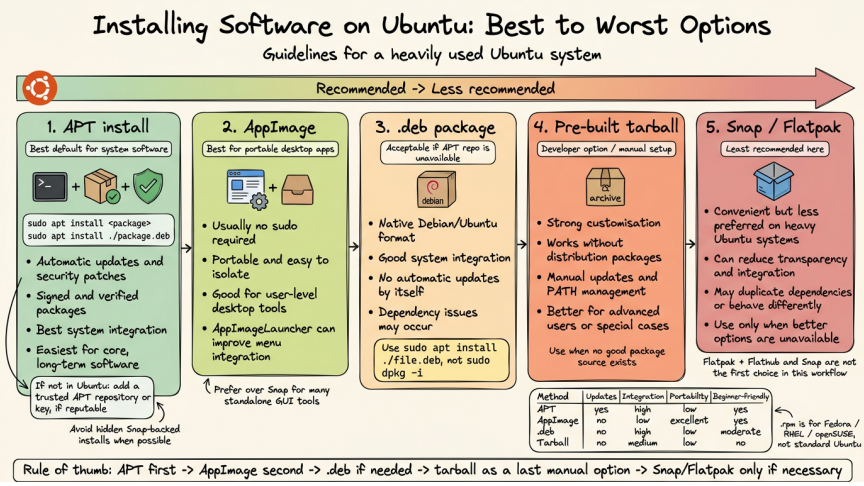


Figure 2.4: Guidelines for installing software on Ubuntu

> `portable`, or not. Portable installation is easier to remove, as it does not leave leftovers in the core system, but does not allow for automatic updates. On the other hand, non-portable applications are usually better for daily/intensive use, but not *sandboxed*, and harder to remove.

2.3.1 Git and GitHub configuration

1. Github often requires double authentication, that you can set on https://github.com/settings/security. Recommended to use *passkey* if supported by your device (unfortunately, Linux OS usually do not support it), or use authenticator app, such as Microsoft Authentificator. **Deterred against the use of SMS + mobile phones**.

2. Configure your access to Github:

i. Choose either SSH, or HTTPs.
ii. For HTTPS, configure one token per device, under https://github.com/settings/tokens. Give the token the name of your laptop, its configuration and originals owner. Assign it at least the repo and read scopes. Copy it once -> **it won't be available afterwards!!**.

3. General options to configure:

i. Use either the command line (create also automatically the file `~/.gitconfig`):

```
git config --global user.name "bastienchassagnol"
git config --global user.email "bastien.chassagnol@univ-amu.fr"
```

[8] `apt-get` works similarly, but is now deprecated
[8] Fetch the latest stable archive, extract it, and add executables to your `PATH`, see example under Section 2.3.12

ii. Or directly, using text editor, file `~/.gitconfig` (more error-prone to indentation errors). Find my default git config file here (for comprehensive description of options, report to Formation Git 2024, a tutorial by Benjamin):

```
[user]
  name = bastienchassagnol
  email = bastien.chassagnol@univ-amu.fr
## modern default git options
[init]
  defaultBranch = main

## simplifying solving git conflicts + troubleshoot solving
[merge]
  conflictstyle = zdiff3
[rerere]
  enabled = true

## tolerate distinct end of lines
[core]
  autocrlf = input

## visualisation
[tag]
  sort = version:refname
[branch]
  sort = -committerdate
[credential]
  helper = store
```

4. Avoid typing your password for every login:

i. For HTTPS + PAT token protocol, use `git config --global credential.helper store` for **permanent** registration of the pass-

word (should we not encrypt the file??), or `git config --global credential.helper 'cache --timeout=3600'` for temporary access[9].

ii. Define SSH key.

2.3.2 AppImageLauncher

```
## download the .deb archive, with the proper architecture configuration
## https://github.com/TheAssassin/AppImageLauncher/releases/tag/v3.0.0-beta-3

sudo apt install appimagelauncher_3.0.0-beta-2-gha287.96cb937_amd64.deb

## update dependencies
sudo apt update
sudo apt install -f
```

- For deleting apps installed with AppImage Launcher, you'll need to delete both *Applications* folder, and `~/.local/share/applications/` that stores executables.

2.3.3 PDF management

2.3.3.1 Sioyek as PDF Viewer

`Sioyek` is an open-source PDF viewer focusing on technical papers with mathematical formulas. Major uses cases are coloured *highlights* of text sections with `h` shortcut, smart jump to figures by clicking on them, and bookmark text sections with shortcut `b`.

1. Choose between portable (for isolated environment), or standard distribution under Sioyek releases

[9]Time-out units are reported in seconds, accordingly, this amounts to one hour

2. Unzip the archive, within it, make it executable with `chmod +x`.
3. Run it

> ℹ A step-bystep tutorial for the installation from source an Ubuntu application, from downloading the archive to adding it to the Path, and to the Start program

To turn any exec into a **desktop application** on Ubuntu-GNOME, follow this steps:

1. Relocate everything under **`/opt/sioyek`**[10]. **Note that the structure of a package may not follow exactly this of `sioyek`!!

```
## create required folders

sudo mkdir -p /opt/sioyek
sudo mkdir -p /etc/sioyek
sudo mkdir -p /usr/share/sioyek

## relocate, if not already done, the sioyek builds

### core executables
sudo cp ~/sioyek/build/sioyek /opt/sioyek/
sudo chmod +x /opt/sioyek/sioyek

### confgiuration files
sudo cp ~/sioyek/build/prefs.config /etc/sioyek/
sudo cp ~/sioyek/build/keys.config  /etc/sioyek/
sudo cp -r ~/sioyek/build/shaders /usr/share/sioyek/
sudo cp ~/sioyek/build/tutorial.pdf /usr/share/sioyek/
```

3. Add a distinctive icon, and customise the native Ubuntu launcher ‘Gnome’

```
sudo mkdir -p /usr/share/icons/hicolor/256x256/apps ## folder to store the siokey.p

sudo tee /usr/share/applications/sioyek.desktop > /dev/null <<'EOF'
[Desktop Entry]
Version=1.0
Type=Application
Name=Sioyek
GenericName=PDF Viewer
Comment=Fast keyboard-driven PDF viewer
Exec=/opt/sioyek/sioyek %path to the sioyek exe
Icon=sioyek
Terminal=false
Categories=Office;Viewer;
MimeType=application/pdf;
StartupNotify=true
EOF

## Refresh desktop database
sudo update-desktop-database
```

4. (Optional) Make Sioyek the default **PDF viewer**, using either:
    a. GUI method:
        1. Right-click any PDF
        2. **Properties → Open With**
        3. Select **Sioyek**
        4. Click **Set as default**
    b. or the command line:

```
xdg-mime default sioyek.desktop application/pdf

## to check default
xdg-mime query default application/pdf
```

5. add to the Path, creating a symbolic link

```
sudo ln -s /opt/sioyek/sioyek /usr/local/bin/sioyek
```

2.3.3.2 Sejda for editing PDFs in an interactive way

- Web version
- Sedja *add-on* for Chrome

2.3.3.3 PDFtk

- To edit PDFs, if you prefer the command line.

2.3.4 Image Editors (to replace Paint on Windows)

Sorted from closest experience to Paint, to most generalist:

1. `Drawing`

i. Add Drawing to the `apt` repository of allowed PPAs: `sudo add-apt-repository ppa:cartes/drawing`
ii. Install running `sudo apt install drawing`

2. `Libre Office Draw`
3. `Gimp` (closer equivalent to `Photoshop`):

```
sudo apt install gimp
sudo apt install gimp-plugin-registry ## add extensions
```

4. `Inkscape` for generating **vectorial**, and fully deformable icons

[10] default repo for third-party apps, preventing conflicts with distributed, native packages

```
sudo apt update
sudo apt install inkscape
```

2.3.5 LibreOffice

1. Install main software (such as Libre Office Write, or Calc) with `sudo apt install libreoffice libreoffice-gtk3`
2. Relevant extensions, that you can add using `Tools -> Extensions`

i. Rotate Images in free text documents with *Rotate* extension
ii. Install dictionaries + extension Grammalecte and LanguageTool[11]. *needs to check if LanguageTool is truly relevant, is AI-based, but also resource-consuming.*

```
## French dictionary
sudo apt update
sudo apt install libreoffice-l10n-fr ## add French hyphenation
```

iii. Install Latex extension

3. Fonts customisation:

i. Install core Windows fonts:

```
sudo apt update
sudo apt install ttf-mscorefonts-installer ## Install Microsoft core fonts
fc-cache -f -v ## rebuild font cache, such that the fonts now appear In Libre Office
```

[11] You may configure the Java version used by default under `Tools -> Options -> Advanced`

ii. Extra fonts installation[12]. Most popular fonts can be found under Every-Font GH repository:

```
mkdir -p ~/.local/share/fonts ## ensure personal fonts folder is created
wget -P ~/.local/share/fonts "https://raw.githubusercontent.com/serendipious/every-font/master/Century
fc-cache -f -v ## update font cache
```

Core typographic options

The main stylistic variants of fonts you can play with are:

1. Weight (Light, Regular, Bold, …)
2. Angle or shape of the letters (regular versus italic)
3. Width
4. Serif vs Sans-Serif. This property is structural to a font, for instance, `Times New Roman`is Serif, and `Century Gothic` is `Sans-serif`. **For data visualizations, or presentations, `Sans-serif` typefaces are favoured, while it's the reverse for printed documents.**
5. Advanced font options and customisations, such as controlling for *ligatures* connecting two letters, can be accessed in `LibreOffice → Format → Character → Features` (see Figure 2.5):

   Make sure, if avalaible, to download for each font, the regular, bold, italic-bold, and italic versions at the very least.

---

[12] Alternatively, you also have the native Fonts manager tool under GNOME launch start

Figure 2.5: Main typographic options

2.3.6 Docker

1. Download dependencies, configuration files and add repository to apt for automatic updates:

```
sudo apt update
sudo apt install ca-certificates curl ## dependencies for downloading packages, and verifying HTTPS pr

## Part I: create, download and make Docker apt key readable for verifying its integrity
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

## Part II: define where, and which version of Docker, should be donwloaded + how to verify its integ
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc ## folder previously defined
EOF

## Part III: update with newer configuration
sudo apt update
```

2. Install Docker toolkit with `sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin`
3. Run Docker without `sudo`rights for the main laptop user:

```
# create a Unix group allowing Docker to interact with system-level resources without admin rights
sudo groupadd docker
# and add curent user of the session to it
sudo usermod -aG docker $USER
```

4. Check installation with `sudo docker run hello-world`. If not running by default upon login, enable automated start with `sudo systemctl start docker`. More details here.

   Tutorials for Docker

- Benjamin presentation
- VsCode, R and Docker with DevContainers
- VsCode, Python and Docker with DevContainers
- DevContainers, from INRIA trainings

2.3.7 Teams

- **Use the web version, as Microsoft no longer supports .deb versions for Linux.**

2.3.8 Web browsers

2.3.8.1 Firefox

> **Avoid running directly `sudo apt install firefox` -> it installs by default `snapd`, and will use not stable, beta `Snap` version of Firefox.**

1. Import, and add Mozilla APT signing key:

```
## Download APT key
wget -q https://packages.mozilla.org/apt/repo-signing-key.gpg -O- | sudo tee /etc/apt

## Add it to the Apt registry
cat <<EOF | sudo tee /etc/apt/sources.list.d/mozilla.sources
Types: deb
URIs: https://packages.mozilla.org/apt
```

```
Suites: mozilla
Components: main
Signed-By: /etc/apt/keyrings/packages.mozilla.org.asc
EOF

## Provide higher priority to Mozilla Firefox

echo '
Package: *
Pin: origin packages.mozilla.org
Pin-Priority: 1000
' | sudo tee /etc/apt/preferences.d/mozilla
```

2. Install Firefox using Mozilla provider:

```
sudo apt update
sudo apt install firefox

which firefox ## should not report `snapd` folder
apt policy firefox
```

2.3.8.2 Google Chrome

1. Install Google Chrome, by downloading the .deb archive, then run it:

```
sudo apt install google-chrome-stable_current_amd64.deb`
```

2. Download, and add Google Chrome APT signing key, for automatic updates:

```
## Download APT key
wget -q -O - https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -

## Add it to the Apt registry
echo "deb [arch=amd64 signed-by=/usr/share/keyrings/google-linux-signing-key.gpg] htt

## Update to the most recent version
sudo apt update
```

3. add the *Grammarly* Chrome plug-in extension for automatic checking of the spelling, and grammar (choose one specific regional English, and stick to it)

2.3.9 Slack

1. Download Slack application
2. Run it with `sudo apt install slack-desktop-4.47.69-amd64.deb`
3. Update with:

```
sudo apt-get update
sudo apt-get upgrade slack-desktop
```

2.3.10 Zulip

1. Run following instructions:

```
sudo apt install curl ## if not already installed
sudo curl -fL -o /etc/apt/trusted.gpg.d/zulip-desktop.asc \
    https://download.zulip.com/desktop/apt/zulip-desktop.asc
echo "deb https://download.zulip.com/desktop/apt stable main" | \
```

```
    sudo tee /etc/apt/sources.list.d/zulip-desktop.list
sudo apt update
sudo apt install zulip
```

2. Log-in (recommended with your Github profile) on each of the channels you want to connect, to keep notified about newest posts.

2.3.11 Cytoscape

Major default: only supported by Java version 17 (and not by more recent, and stable Java versions!!)

1. Install Java version 17 with `sudo apt install openjdk-17-jdk`[13]
2. Follow this tutorial

i. Retrieve the latest version
ii. Make it executable with `chmod u+x <filename.sh>`
iii. Run with `bash <filename.sh>`
iv. Launch with `Cytoscape &`.

2.3.12 Zotero

1. `sudo apt install libreoffice-java-common -y` adds Java dependencies to seamlessly integrate Zotero with Libre Office *plug-in* (for instance, automated retrieval of bibliographic references).
2. Retrieve the latest stable version on https://www.zotero.org/download/[14].
3. Run following instructions + Synchronise by installing Zotero Connector on your website:

[13] On Ubuntu, a more modern version of Java, the 21, was installed
[14] Alternatively, download with `curl`

```
#!/bin/bash

## part I: extract archive
sudo tar -xjf zotero.tar.bz2 -C /opt
sudo mv /opt/Zotero_linux-x86_64 /opt/zotero

## Part II: add the executable
sudo ln -s /opt/zotero/zotero /usr/local/bin/zotero
```

4. (Optional: add a custom Desktop, and launcher icon for an executable) Create file `~/.local/share/applications/zotero.desktop`, with the following content. Make the launcher executable with `chmod +x ~/.local/share/applications/zotero.desktop`

```
[Desktop Entry]
Name=Zotero
Exec=/usr/local/bin/zotero
Icon=/opt/zotero/icons/icon128.png
StartupWMClass=Zotero
Type=Application
Terminal=false
Categories=Office;
MimeType=text/plain;x-scheme-handler/zotero;application/x-research-info-systems;text/
X-GNOME-SingleWindow=true
```

5. Inspired from Zotero hacks tutorial, enable unlimited synced storage for articles on multiple machines.

i. Add required plug-ins, `Better BibTeX` (mandatory for custom expert as a plain .bib file) + `Zotmoov`, which replaces previous extension `Zotfile` for classifying articles per author's name.

ii. Edit Zotero preferences[15]: - Uncheck the option to create automatic web page snapshots (increases cluttering with lots of small files added) - Uncheck full-text sync - Set apart in `Edit->Settings->Advanced` the

- Base directory (should be the folder synced by your File sharing software), with
- Custom Data directory location. This local folder should be located in a different position, and will be managed by Zotero itself

```
![Zotero Database Configuration](zotero-advanced-configuration.png)
```

iii. Edit `Better BixTex` *plug-in*: - Personally, use `[auth:lower][year][journal:lower:abbr]` - Should be the same across machines.

2.3.13 Quarto, R and RStudio

2.3.13.1 R, and recommended packages

1. To get the latest R Versions, add the signing key, along with the repository:

```
sudo mkdir -p /etc/apt/keyrings
sudo wget -O /etc/apt/keyrings/cran.gpg https://cloud.r-project.org/bin/linux/ubuntu/marutter_pubkey.a
echo "deb [signed-by=/etc/apt/keyrings/cran.gpg] https://cloud.r-project.org/bin/linux/ubuntu jammy-cr
```

2. Install R:

i. Install basic dependencies

[15]What matters the most here is uniformity across platforms

```
sudo apt update
sudo apt install r-base r-base-dev -y
R --version
```

ii. Install recommended Ubuntu system libraries for scientific computing and/or connection:

```
## Compilation tool
sudo apt install cmake

Network, imaging, and plotting libraries
sudo apt install \
build-essential \
libpng-dev \
libjpeg-dev \
libtiff-dev \
libcairo2-dev \
libcurl4-openssl-dev \
libssl-dev \
libxml2-dev \
zlib1g-dev

## scientificic libraries
sudo apt install \
  build-essential \
  gfortran \
  libblas-dev \
  liblapack-dev \
  libgsl-dev
```

3. Configure default CRAN mirror.

i. default choice is https://cloud.r-project.org, as the official CRAN miror, and the most up-to-date
ii. For Ubuntu, the **Posit Package Manager** mirror, by providing prebuilt Linux binaries, is much faster to compile. To install it, modify the local `~/.Rprofile` file:

```
echo 'options(repos = c(CRAN = "https://packagemanager.posit.co/cran/__linux__/<ubuntu-version>/latest
```

4. Install your packages, by organising them as a package repository with dependencies. Notably, install those of Stephen Turner, as essential to most bioinformatic pipelines: `remotes::install_github("stephenturner/Tverse", upgrade=TRUE)`.

2.3.13.2 RStudio (Desktop IDE)

1. Go to: https://www.rstudio.com/products/rstudio/download/#download/ Choose the **Ubuntu/Debian `.deb` package** for your architecture (amd64).
2. Install the `.deb` package:

```
sudo apt install ./rstudio-2025.09.0-422-amd64.deb
```

3. Install Air, as the recommended, fast and modern formatter for R code. Then, customise your RStudio settings, explicitly reporting the path location of Air formatter, see Tip 2.

```
uv tool install air-formatter
```

Tip 2: Sync RStudio options across platforms

1. **Locate the repository on your system that stores RStudio settings**. There's no direct means for exporting RStudio parameters that have been modified by the user. Instead, the key directory storing core RStudio settings is located under `~/.config/rstudio/` for Linux, and `%AppData%\RStudio` for Windows[16]. The key files in this repository are:

File	Purpose
`rstudio-prefs.json`	Global preferences modified by the user
`keybindings/`	Custom keyboard shortcuts
`themes/`	Custom themes (if any)
`snippets/`	Code snippets
`dictionaries/`	Custom dictionaries beyond English, and French

2. **Identify what it's worth relevant for syncing**: indeed, while you could export all the RStudio settings for a complete backup, most of them are optional, or barely modified by regular users. Instead, I would identify, and retrieve only the settings explicitly changed by the user. > Would use local, or online diff editors, such as diffcheckers to quickly identify what changed between personal configurations. > Specifically, all external tools that complement the functionalities of RStudio, such as pdf previewer, or external formatters such as Air, are hard-coded as absolute paths, and are accordingly highly specific to a personal laptop.

3. **Homogenise settings across machines**: once you've finalised the customisation of the RStudio settings to your specific OS configuration, simply overwrite pre-existing *user-specific* settings file `rstudio-prefs.json`. **Restart RStudio** to check if changes were properly integrated into the graphical interface[17].

   To simplify the syncing across machines (especially for teams with shared computing environment), use either: 1. Git-based sync, storing your dotfiles. This is the recommended approach. You can find my settings files under `rstudio-prefs.json`. See also Figure 2.6 for a more comprehensive explantion of core changes brought to a default, native RStudio configuration. 2. use symbolic links, or cloud sync (but you have fewer options to adjust for changes)

(a) In yellow, core reproducibility options. Notably, avoid loading your R session, and objects on start, for reduced use of memory

(b) Styling and linting of R code with `Air Posit`, and integration with AI helpers.

Figure 2.6: Core options to change the default behaviour of RStudio, enabling enhanced compatibility across projects, R Versions, and OS platforms.

Controversial claim: use VScode for R development if working inside a Docker container, or on a server that does not provide a RStudio interface, otherwise, it's overkilled. Notably, for R package development, that should be containerised, RStudio is way more flexible

2.3.13.3 Quarto

Go to https://quarto.org/docs/get-started/

[17] `%AppData%` is a shortened alias for this expanded path on Windows: `C:\Users\<YourUser>\AppData\Roaming\RStudio`

[17] If some of the setting changes were not applied, you can usually access them from the Menu options with `Tools > Global Options...`

```
sudo apt install ./quarto-1.3.496-linux-amd64.deb
quarto check
```

$$a = 10$$

2.3.13.4 Cursor with R

1. Install the following R packages:

```
install.packages("remotes")
library(languageserver)
library(rmarkdown)
library(vscDebugger)
library(httpgd)
```

2. (optional, but recommended) Use `radian`as enhanced R terminal with `uv tool install radian`

3. Install the `vscDebugger` extension for VS Code, and run the debug demo script to test it.

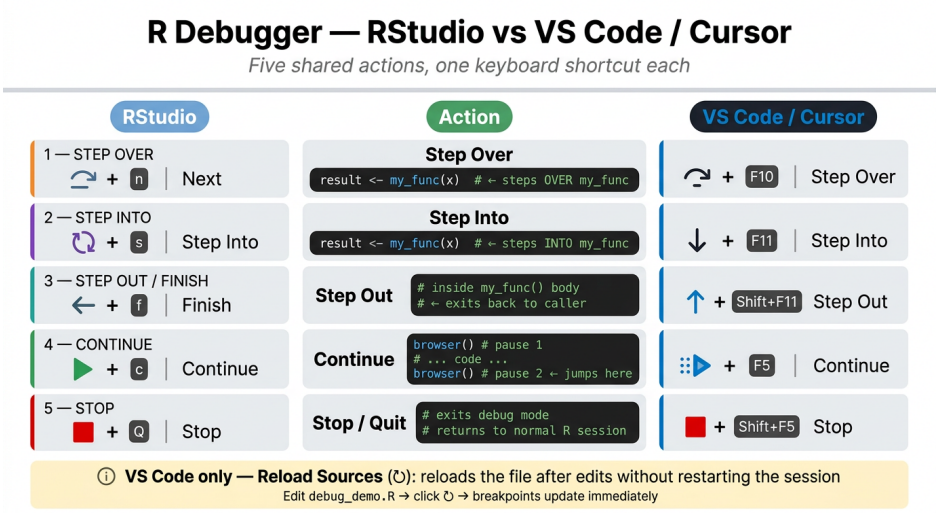


Figure 2.7: Side by side comparison of the debug demo script, and the debugger in action

2.3.14 Python

From **PEP 668**. guidelines, applied to modern Debian/Ubuntu distributions with native Python installed above the `3.11/3.12` version, `pip` is prevented from modifying the **system Python** as it could interfere with te core installation program `apt`.

This section is certainly the most controversial of this whole Ubuntu tutorial, as guidelines for a modern programmatic use of Python evolve rapidly, without reaching a full agreement

2.3.14.1 Python tools

Modern Python development guidelines

1. The latest versions of Ubuntu, including `Noble`, strongly deter from installing system-wide versions of recent Python packages. In other words, avoid running `sudo apt install python3.14-full`, and updating Deadsnakes PPA.
2. `uv` fully replaces the functionalities and features brought by a variety of slower, standalone Python tools (`pip`, `poetry`, `virtualenv`, `pipx`, …), providing a faster, versatile and unified framework, see Figure **??**. **The features provided by `poetry` for package development may still not be fully covered by `uv`, and `pip`remains the legacy solution in older HPC systems:**

```
/usr/bin ### Pre-installed Ubuntu system
 └── system Python (used by OS)

uv
 ├── managed Python versions
 ├── CLI tools (black, ruff, jupyterlab)
 └── project environments

~/.local/share/uv/python/
   ├── python3.14
```

3. `conda`, and even `mamba` have been outdated by `pixi` for multi-language projects. **`uv` is definitely the go-to solution for Python-centric projects, but does not support yet cross-language project managements**. Of note, `uv` natively enforces the use of virtual environments when running Python scripts, see Table **??**.

4. The field renews fast -> stay alert, and up-to-date to the latest trends in this tech world.

Feature	uv	Poetry	Pixi
Speed			
Dependency management			
Virtual environments			
Python version install			
Package publishing	Acceptable	Best	Limited
Non-Python dependencies			
CI performance			
Maturity	Medium	High	Medium

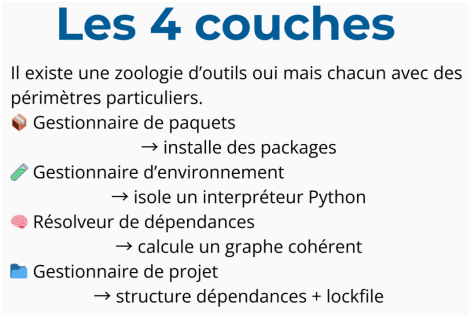
2.3.14.2 One Tool to bring them all and in the jungle of Python tool, bind them all: a `uv` story plot

I now describe the core steps to use the latest stable Python version (currently, 3.14), without breaking core Linux dependencies using pre-installed Python version.

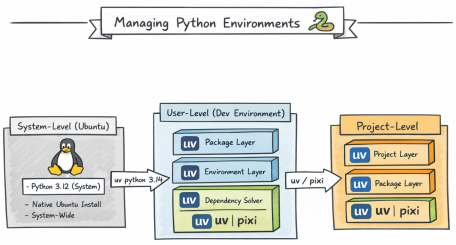
1. Install `uv` with the following command: `curl -LsSf https://astral.sh/uv/install.sh | sh`.
2. Install latest Python version with `uv`: `uv python install 3.14`.
3. Two configuration levels for setting up the Python version

   i. At the project level, run `uv python pin 3.14`.
   ii. At the user level: `bash cd ~ echo "3.14" > ~/.python-version`

4. Use `uv` instead of `pipx` to install core Python tools for package dependency, web IDEs, modern Python formatting and styling, ….
   As `pipx`, all these tools are automatically added to the user's `PATH`:



(a) Each tool used to have its perimeter hence the diversity of Python execs

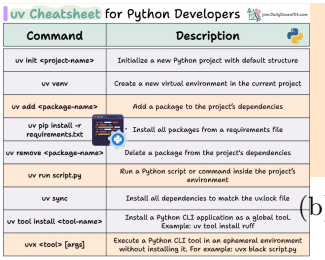


(b) The three levels of Python environments, from system-wide (not to be modified) to the project level

	Paquets	Env	Resolver	Projet
pip	✓	✗	✓	✗
venv	✗	✓	✗	✗
conda	✓	✓	✓	✗
mamba	✓	✓	✓ (rapide)	✗
poetry	✓	✓	✓	✓
uv	✓	✓	✓ (rapide)	✓
pixi	✓	✓	✓	✓

(c) Side-by-side compariosn of the core use cases of each Python tool -> uv supersedes them all.

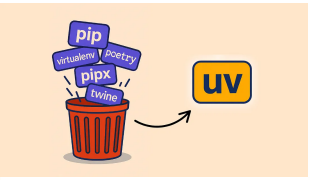
Figure 2.8: Comparison of Python managers.



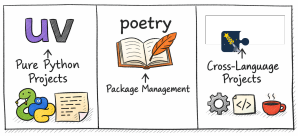
uv Cheatsheet for Python Developers

Command	Description
uv init <project-name>	Initialize a new Python project with default structure
uv venv	Create a new virtual environment in the current project
uv add <package-name>	Add a package to the project's dependencies
uv pip install -r requirements.txt	Install all packages from a requirements file
uv remove <package-name>	Delete a package from the project's dependencies
uv run script.py	Run a Python script or command inside the project's environment
uv sync	Install all dependencies to match the uv.lock file
uv tool install <tool-name>	Install a Python CLI application as a global tool. Example: uv tool install ruff
uvx <tool> [args]	Execute a Python CLI tool in an ephemeral environment without installing it. For example: uvx black script.py

(a) Main commands to replace old-fahsioned Python tools by `uv`.



(b) Caricature illustrating the recent domination of `uv`



(c) Decision tree for the choice of Python tool, depending on the scope: `uv` for pure Python projects (for instance, running bioinformatic analyses), `poetry` (and `uv` with less support) for Python packages, and `pixi` for cross-language projects. **Note that both `conda` and `mamba` ar eno longer required in the modern Programmer stack.

```
uv tool install ruff ## ~ equivalent to pipx install ruff
uv tool install "dvc[ssh]" # recommended data management tool, and versining, if exceedging git-lfs
uv tool install black
uv tool install jupyterlab ## `JupyterLab` outdates previous `JupterNotebook` as a web-based, and user
uv tool install ipython
uv tool install pre-commit
uv tool install poetry
uv tool install cookiecutter `Cookiecutter` generates Python templates for modern packaging.
```

Python version management: core checks

- Check integrity of system Python:

```
ls /usr/bin/python* ## -> ## only one version expected, Python 3.12 with modern Linux
/usr/bin/python3.12
```

- Check version used by `uv`:

```
ls -la ~/.local/share/uv/python* ## path returned should be located within the HOME directory
```

Checking current Python version used:

- From the terminal (system-wide), `ls -d /usr/lib/python3.*`, or `which -a python` should only return a unique version of Python, the one used by Ubuntu.
- From a Python shell (including Jupyter Lab):

```
import sys
print(sys.version)
print(sys.executable)
print(sys.path)
```

Relevant sources for Python tools

- From Pip to Lightning: How uv Opened My Eyes to Better Python Workflows
- LinkedIn portfolio on Python dependencies management tools

2.3.14.3 Marimo to replace Jupyter-Lab

1. Install Marimo, including commonly used tools: duckdb (SQL cells), polars/pyarrow (SQL output), ruff (formatting), vegafusion (charts)

```
uv tool install "marimo[recommended]"

# Verify
marimo --version
```

2. Add Cursor/VS Code extension:

```
cursor --install-extension marimo-team.vscode-marimo
```

3. Commonly used keybindings to be added:

```
  // --- □ Marimo ---
  // Open current .py file as a marimo notebook in the editor
  {
    "key": "ctrl+alt+m",
    "command": "marimo.openNotebook",
    "when": "resourceExtname == '.py'"
  },
  // Create a new sandboxed marimo notebook via terminal
  {
```

```
    "key": "ctrl+alt+shift+m",
    "command": "workbench.action.terminal.sendSequence",
    "args": { "text": "uvx marimo edit --sandbox \n" }
  },
  // Run current marimo notebook as a read-only app
  {
    "key": "ctrl+alt+r",
    "command": "workbench.action.terminal.sendSequence",
    "when": "resourceExtname == '.py'",
    "args": { "text": "uvx marimo run '${file}'\n" }
  }
```

4. Interactive example is provided in demo.py.

2.3.15 Nextflow as workflow management tool

Installation process as standalone, and automatically up-to-date tool. µµCheck first that Java version at least 17 is installed, see Section 2.3.11:

```
curl -s https://get.nextflow.io | bash
chmod +x nextflow # make nextflow executable

# move nextflow into execubtale path
sudo mv nextflow $HOME/.local/bin/
```

2.3.16 Latex

```
sudo apt install texlive-full -y
## other commonly used packages
sudo apt install texlive-latex-extra texlive-fonts-recommended texlive-science biber latexmk -y
```

2.3.17 WhatsApp

- No supported local Linux installation
- Instead, recommended to use the Web version of WhatsApp, then pair it with your mobile device.

2.3.18 Cursor

1. Download DEB extension.
2. Run it:

```
sudo apt install ./cursor_2.0.77_amd64.deb
```

3. Sync your Cursor configuration across devices, see Tip 3:

> Tip 3: Export Cursor settings
>
> 1. Export the core settings:
>
> ```
> i.  Open Command Palette (`Ctrl/Cmd + Shift + P`), then search: **“Preferences: Ope
> ii. Export the file in your new device
> iii. Default Linux installation: `~/.config/Cursor/User/settings.json`
> ```
>
> 2. Export the keybinding shortcuts:
>
> i. Open: **“Preferences: Open Keyboard Shortcuts (JSON)”**
> ii. Export `keybindings.json` into your local folder: `~/.config/Cursor/User/keybindings.json`
>
> 3. (**the main difference with VSCode**). Contrary to `VSCode`, you need to export the list of extensions manually:
>
> i. Retrieve the list of extensions running `cursor --list-extensions > extensions.txt`

ii. Install them back with:

```
grep -v '^\s*#' "~/.config/Cursor/User/extensions.txt" \
| grep -v '^\s*$' \
| xargs -I{} cursor --install-extension {}
```

4. The whole list of extensions, and Cursor settings are available under the Cursor config folder.

   More details are provide in *Sync Cursor Settings the Dotfiles Way* website, including the symlink commands tu fully automate this process

   The harcore way, for syncing configuration files across numerous users, cf Benjamin, and Cursor Config Sync

2.3.19 Password Manager for Linux -> BitWarden

'BitWarden is the recommended password manager for Linux systems for a variety of reasons: it's mostly free, compliant with most web browsers (Firefox, Chrome, Opera, ...), including native Google password manager, and can sync your password database over an endless number of devices. There's even a lightweight application for Iphone, and Android mobile phones. Besides, the storage expends beyond raw passwords, ranging from passkeys, to SSH secrets, or credit card information. Finally, it's compatible with most OS, including Windows and Linux.

Accordingly, all these features fax expand the facilities delivered by native Ubuntu Gnome `Passwords and Keys` application manager, which besides can only be installed locally.

One major drawback, though, is that it does not check for duplicated values in an automated way.

Follow the following steps for its installation[18]

```
## Install Flatpak, and Gnome Flatpak plugin
sudo apt install flatpak
sudo apt install gnome-software-plugin-flatpak

## Add Flatbu repository for software storage
flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpa

## Install and run BitWarden locally
flatpak install flathub com.bitwarden.desktop
flatpak run com.bitwarden.desktop

## Install Web Add-on, going to https://chromewebstore.google.com/detail/bitwarden-pa

## Export, then import your password database from other web browsers
```

In the `Auto-fill` tab of the Parameters configuration, leave unticked the option `Do not auto-fill on page load` + choose `Default URI detection` option as `Exact` to avoid ambiguities. Counterintuitively, go to Web Vault BitWarden to bulk remove passwords. This option is not proposed by default in the regular BitWarden.

2.4 Core numerical tools provided by AMU

List of numerical tools used in Amu

[18] Contrary to best guidelines suggested in Note 1, only the `flatpak` and `snap` installation are supported for Ubuntu devices -> no automatic updates with native `apt` Ubuntu dependency manager

2.4.1 AMUZoom

1. Retrieve latest version, choosing proper Linux distributions and OS architecture under Zoom downloads. Also avalaible online. Strongly recommended to install Zoom locally.
2. Run `sudo apt install ./zoom_amd64.deb`
3. **Log-in using your university email address, using `SSO`.** The account is provided with unlimited meetings available by default, use following domain: `univ-amu-fr`. [19]

   2 elements, listed below, to keep in in mind, when creating a Zoom meeting to enable any external user, with the password, to ling-in, and share its screen, see Figure 2.10:

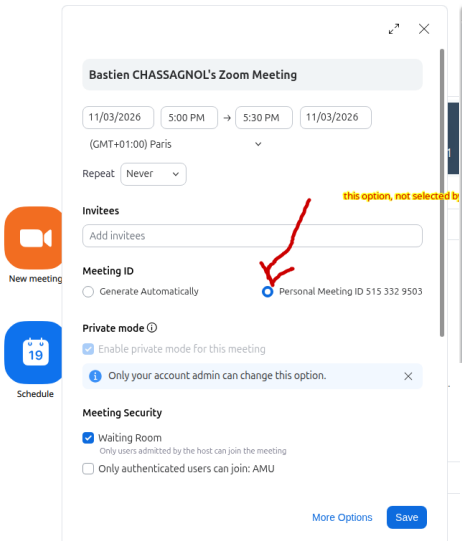
To go further, check the following online resources:

- Guidelines for Zoom Meeting creation, and management
- Video tutorial
- Guidelines for students

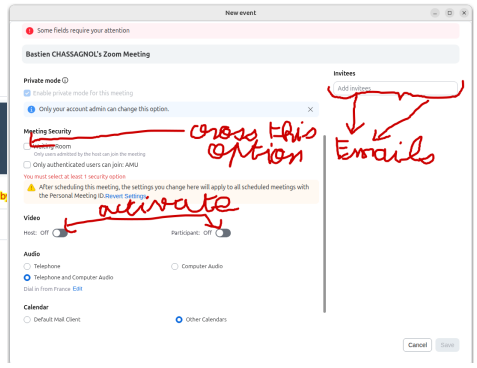
2.4.2 WooClap for QCMs

Go to Wooclap, and use your **instutional mail adress** for single sign-on across platforms.

[19] Alternatively, use the web version available here



(a) Zoom first login



(b) Zoom second step of login

Figure 2.10: Be careful about selecting the second, non-default option, if you want to allow external suers without any Amu mail address to connect on your Zoom session.

2.5 Working with a remote server

Retrive its identity on the server

```
id ${USER}
uid=1001(bastien) gid=1001(bastien) groups=1001(bastien),4(adm),24(cdrom),27(sudo),30(dip),46(plugd
```

- list of groups on the server, identify is defined by the combination of an username, `UID` (user ID), and `GID` (group ID) on both servers.

```
id ${USER}
uid=1007(bastienc) gid=1007(bastienc) groupes=1007(bastienc),1043(dtoo_project)
```

More information avalaible under Server Management tutorial.

2.5.1 Starting configuration

```
# ~/.bashrc (minimal and robust)

# Local user binaries first
export PATH="$HOME/.local/bin:$PATH"
export PATH="$PATH:$HOME/scripts/tools"

# Auto-switch interactive Bash sessions to user-local Zsh
if [ -x "$HOME/.local/bin/zsh" ] && [ -n "$PS1" ] && [ -z "$ZSH_VERSION" ]; then
  exec "$HOME/.local/bin/zsh" -l
fi
```

```
# ------------------------------------------------------------------
# Dotfiles migration via SCP (run these on your LOCAL machine)
# ------------------------------------------------------------------
# 1) Create target directories on server
# ssh mmg-server 'mkdir -p ~/.ssh'

# 2) Copy core dotfiles
# scp ~/.bashrc ~/.gitconfig mmg-server:~/

# 3) Copy SSH config
# scp ~/.ssh/config mmg-server:~/.ssh/config

# 4) Secure SSH permissions on server
# ssh mmg-server 'chmod 700 ~/.ssh && chmod 600 ~/.ssh/config'

# Optional: copy SSH keys (only if you explicitly want this)
# scp ~/.ssh/id_ed25519 ~/.ssh/id_ed25519.pub mmg-server:~/.ssh/
# ssh mmg-server 'chmod 600 ~/.ssh/id_ed25519 && chmod 644 ~/.ssh/id_ed25519.pub'
``


### SSH configuration

> Objective is dual: avoid typing your password each time you're connecting to a remo

1.  Set up SSH remote access (with key-based auth)

``` bash
## remote server must have openssh-server installed
sudo apt install openssh-server
```

2. Generate a pair of public-private SSH key on your local machine[20]

[20]Using a passphrase is not compuslory, but safer
```

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

This creates:

```
- `~/.ssh/id_ed25519` (private key)
- `~/.ssh/id_ed25519.pub` (public key)
```

3. Copy your key to the server with `ssh-copy-id username@server_ip`. You'll no longer need typing your password once typed once.
4. (Optional) Keep under `~/.ssh/config` file the key features describing each remote cluster. For instance, my configuration for the IFB is the following, and to connect to it, I only need typing now `ssh ifb-core`. A second advantage of using this format for storing hosts is that it can be readily used by `VS Code SSH Remote` connection extensions.

```
## □ alias for the host connection, either "ifb-core" or "ifb"
Host ifb-core ifb

  ## □ The real hostname (or IP) of the remote server
  HostName core.cluster.france-bioinformatique.fr

  ## □ Username used on the remote machine
  User bchassagnol

  ## □ Path to your private SSH key
  IdentityFile ~/.ssh/id_ed25519

  ## □ Force SSH to use ONLY the specified key above, avoiding complex managing of key version
  IdentitiesOnly yes

  ## □ Order of authentication methods: SSH key, then keyboard-interactive, then fallback to password
  PreferredAuthentications publickey,keyboard-interactive,password
```

```
  ## □ Automatically add this key to ssh-agent when used
  AddKeysToAgent yes

# Host configuration for the new server
Host mmg-sb-05 new-mmg-cluster
HostName 139.124.156.52
Port 22218 # is the port, avoiding, unlike the popular 22, to be hacked.
User bastienc
IdentityFile ~/.ssh/id_ed25519
IdentitiesOnly yes
PubkeyAuthentication yes
PreferredAuthentications publickey,keyboard-interactive,password
AddKeysToAgent yes
ForwardX11 yes
ForwardX11Trusted yes
```

My host configuration, enabling by default graphical interfaces to run on the remote server, is avalaible under config-ssh

---

Ssh configuration recommendations

On older Ubuntu configurations, the `shh-agent`, and `shh-add` processes are not activated by default on startup. Check this by running `systemctl --user enable ssh-agent`.

Report to Table **??**, and Figure 2.11 for further details

Practice	Why
Prefer Ed25519 keys	Small, fast, modern default.
Use a pass phrase on the key	Protects the key if the disk is copied or stolen.
`~/.ssh/config` per host	Aliases, and login recommendations
Never commit private keys	Keep `id_*` out of git and cloud-synced folders.
`ssh-copy-id` once per host	Standard way to populate `authorized_keys`.

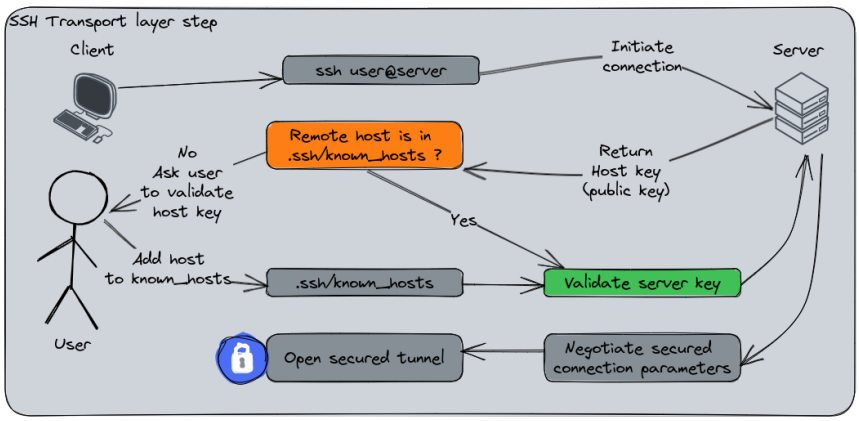


Figure 2.11: Overview of an initial SSH connection

Option	Purpose	Effect on X11
`-X`	X11 forwarding (untrusted)	Runs GUI apps remotely
`-Y`	X11 forwarding (trusted)	Runs all GUI apps remotely
`-C`	Compression	Reduces network usage and can speed up GUI over slow connections

2.5.2 MMG cluster, and recommendations for project management

About projects timeline, and structure, follow these tips as soon as *collaboration* is involved:

- Initiate `git + github` at the real start up of the project
- Use only the `/mnt/<cluster-name>/projects/` folder (no projects in your home, except for a quick run, and test)
- When creating the project, set carefully group and user permissions, following guidlines of document https://github.com/BAUDOTlab/teaching_material/blob/main/Server%20Management/Server_Management.pdf, section **Creating a New Project**
- Complement the `README.md` (to be created) at the root of the folder, with one short line per directory in the `projects` folder quickly summarising the overarching objective of the study

~/remote_project –>

Upon my recent transition from the Windows world to Linux, and huge frustration resulting from spending a lot of time recovering tutorials and disseminated configuration files when configuring a new laptop, here's my detailed tutorial about best practices when starting from fresh on a Ubuntu session: https://baudotlab.github.io/teaching_material/Linux/installation-linux.html (and GH repo: https://github.com/BAUDOTlab/teaching_material) Beyond key commands, I distil some tips to enhance reproducibility across projects, provide default configuration tools for Git, RStudio and VS Code, and suggest software to replace the paying licence of the Office ecosystem. I also tried to concatenate recommendations from the team, including Benjamin, Morgane, Marielle, and philippe notably.

Besides, here are the 3 development guidelines that after discussions with Benjamin, might be the msot controversial, but strongly suggest nonetheless:

- No further support of RServer, nor JupyterLab on the server -> with VS Code, and its extension "Visual Studio Code Remote - SSH", it's easy to automatically retrieve back its local programming framework

  - For Python, switch to VS Code (or Cursor for AI-guided development): `Spyder` is slow, and old-looking, `Jupyter Notebooks` is not reproducible (and limited in functionalities), especially witht he advent of Marimo: https://marimo.io/, `PyCharm`, while offering a slightly better developer experience, has a paying licence that limits its functionalities + all its features are covered by VS Code extensions
  - For R, `RStudio` is definitely the best (to quote Benjamin, "that's the utmost shame that the best IDE had been developed for the worst programming language"). But running it on the server implies strong latency -> suggested workflow: develop, and run your scripts on small examples locally, then scale them on larger datasets using VS Code on the server (or the command line)

- Use `uv` for Python-full projects, `uv` or `poetry` for Python package development, and `pixi` for multi-language projects (see why in following caricature) -> especially, both `conda` and `mamba` are heavy, and way too slow

- About projects timeline, and structure, follow these tips as soon as collaboration is invovled:

  - Initiate `git + github` at the real start up of the project
  - Use only the `/mnt/<cluster-name>/projects/` folder (no projects in your home, except for a quick run, and test)
  - When creating the project, set carefully group and user permissions, following guidlines of document https://github.com/BAUDOTlab/teaching_material/blob/main/Server%20Management/Server_Management.pdf, section **Creating a New Project**

- Complement the `README.md` (to be created) at the root of the folder, with one short line per directory in the `projects` folder quickly summarising the overarching objective of the study

Finally, if you: - Want to collaborate to the tutorial: favour quarto (or notebooks) editable files over PDFs, index your document under `https://github.com/BAUDOTlab/teaching_material/blob/main/_quarto.yml`. You can also report issues directly on the website, clicking on the option "Edit this page" in the footer - For computational nerds, or people generally interested in best programming guidelines, solving computational and dependency nightmares, or simply curious, add a like to this comment -> I'll add a Slack channel to unify debates, and troubleshoot issue solving :-)

3 Modern R Package Development Guide

This guide consolidates package development practice using modern recommendations from the tidyverse ecosystem, especially *R Packages* (2nd edition) by Hadley Wickham and Jenny Bryan.

3.1 Package Structure

A standard R package typically contains:

- `R/`: function source files (one primary function family per file).
- `man/`: generated Rd help files (do not edit manually).
- `DESCRIPTION`: package metadata and dependency declarations.
- `NAMESPACE`: exports and imports, usually generated by roxygen2.
- `tests/testthat/`: automated tests.
- `vignettes/`: long-form user guides.
- `data/`: user-facing datasets (`.rda`), loaded lazily when needed.
- `data-raw/`: scripts that build derived datasets (not shipped to users directly).
- `inst/extdata/`: raw or auxiliary files users may access via `system.file()`.
- `src/`: compiled code (for example C/C++), with `LinkingTo` when relevant.
- `.Rbuildignore`: files excluded from package build.
- `.gitignore`: files ignored by Git.

- `NEWS.md`: notable user-facing changes.
- `README.Rmd` and built `README.md`: project overview and installation instructions.

3.2 Project Setup (End-to-End)

1. Create the package project:

```
usethis::create_package("~/path/to/pkgname")
```

2. Validate naming conventions (letters, numbers, and `.` only; start with a letter).

3. Activate common infrastructure:

```
usethis::use_git()
usethis::use_github()
usethis::use_mit_license("Your Name")
usethis::use_readme_rmd()
usethis::use_news_md()
```

4. Add continuous checks early:

```
usethis::use_testthat(3)
usethis::use_github_action_check_standard()
```

5. During development, use fast feedback loops:

```
devtools::load_all()
devtools::test()
devtools::document()
devtools::check()
```

3.3 Writing Functions

- Put package code in `R/`; avoid `source()` inside a package.
- Do not call `library()` or `require()` in package code.
- Prefer explicit namespace calls (`pkg::fun`) unless a function is imported repeatedly.
- Keep side effects local; for temporary global option/state changes, use `withr`.
- Use snake_case naming and tidyverse style conventions.

3.4 Dependencies and Namespaces

Use `DESCRIPTION` fields correctly:

- `Imports`: packages required at runtime.
- `Suggests`: optional packages used in tests, examples, or vignettes.
- `LinkingTo`: headers needed for compiled code interfaces.
- `SystemRequirements`: non-R system dependencies.

Recommended workflow:

```
usethis::use_package("dplyr", type = "Imports")
usethis::use_package("testthat", type = "Suggests")
```

In roxygen comments:

- `@export` exposes functions to users.
- `@importFrom pkg fun` imports specific functions where justified.

Prefer selective imports over broad `@import` directives.

3.5 Documentation with roxygen2

Document functions directly above their definitions and regenerate docs with:

```
devtools::document()
```

Core tags:

- `@param`: input arguments.
- `@return`: output object.
- `@examples`: runnable examples (skip expensive calls with `\dontrun{}` only when necessary).
- `@seealso`: links to related functions.
- `@family`: groups related functions.
- `@rdname` or `@describeIn`: combine multiple functions on one help page when appropriate.

For package-level documentation, create `R/<pkg>-package.R` with:

- `@keywords internal`
- `@name <pkg>-package`
- a trailing `NULL`

3.6 Testing with testthat

Initialise and run tests:

```
usethis::use_testthat(3)
devtools::test()
```

Guidelines:

- Store tests in `tests/testthat/`.
- Name files as `test-*.R`.
- Group related expectations in `test_that()` blocks.
- Use the most specific matcher available (`expect_identical()`, `expect_equal()`, `expect_match()`, etc.).
- Use snapshot/reference testing where outputs are large or complex.
- Skip selectively (`skip_if_not_installed()`, `skip_on_cran()`) rather than disabling broad test suites.

3.7 Data in Packages

Use the right location for each data type:

- `data/`: user-facing, documented datasets.
- `R/sysdata.rda`: internal package data not exposed to users.
- `inst/extdata/`: external files users can access via `system.file()`.
- `data-raw/`: scripts to build reproducible datasets.

Helpful commands:

```
usethis::use_data_raw()
usethis::use_data(my_dataset, overwrite = TRUE)
usethis::use_data(my_internal_object, internal = TRUE, overwrite = TRUE)
tools::checkRdaFiles()
```

Document datasets with roxygen `@format` and `@source`.

3.8 Build, Check, and Release

Before release:

- Run `devtools::check()` locally and in CI.

- Confirm no unintended notes/warnings/errors.
- Keep `NAMESPACE` generated (do not hand-edit unless absolutely necessary).
- Verify encoding and avoid non-ASCII issues for CRAN where possible.
- Increment versions with semantic intent:

```
usethis::use_version("patch")
```

- Maintain a clear `NEWS.md`.

For CRAN submission, follow current CRAN policies and perform final checks with `devtools::check()` and `devtools::check(cran = TRUE)`.

3.9 Documenting Multiple Functions on One Help Page

Default recommendation: one function family per file and one principal help topic per user concept.

Use:

- `@seealso` and `@family` for cross-links across related pages.
- `@rdname` when several functions should share one help page.
- `@describeIn` for closely related variants/methods with a shared conceptual contract.

Example:

```
#' Basic arithmetic
#'
#' @param x,y Numeric vectors.
#' @name arith
```

```
NULL

#' @rdname arith
add <- function(x, y) x + y

#' @rdname arith
times <- function(x, y) x * y
```

3.10 Parallel Computing in R (Practical Notes)

Check available cores:

```
parallel::detectCores()
```

Local multicore workflows:

- `parallel::mclapply()` (Unix-like systems).
- `foreach` with `%dopar%` plus `doParallel`.
- Use reproducible RNG streams for parallel simulations (for example `RNGkind("L'Ecuyer-CMRG")` with a fixed seed).

Cluster workflows (`PSOCK` clusters), local or remote:

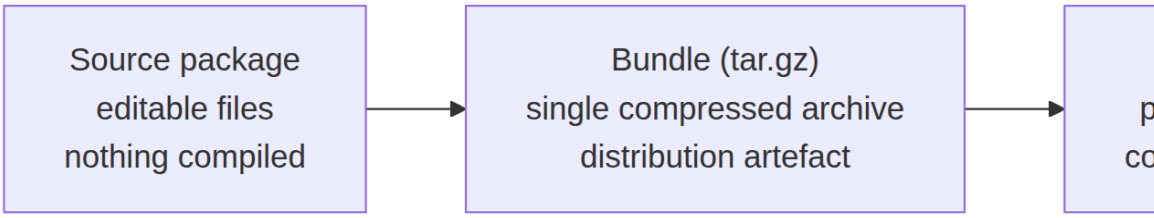
1. Create cluster with `parallel::makeCluster()`.
2. Export required objects via `parallel::clusterExport()` when needed.
3. Evaluate with `parallel::parLapply()` or related functions.
4. Stop cleanly using `parallel::stopCluster()`.

Always close clusters and backend registrations to avoid orphan workers.

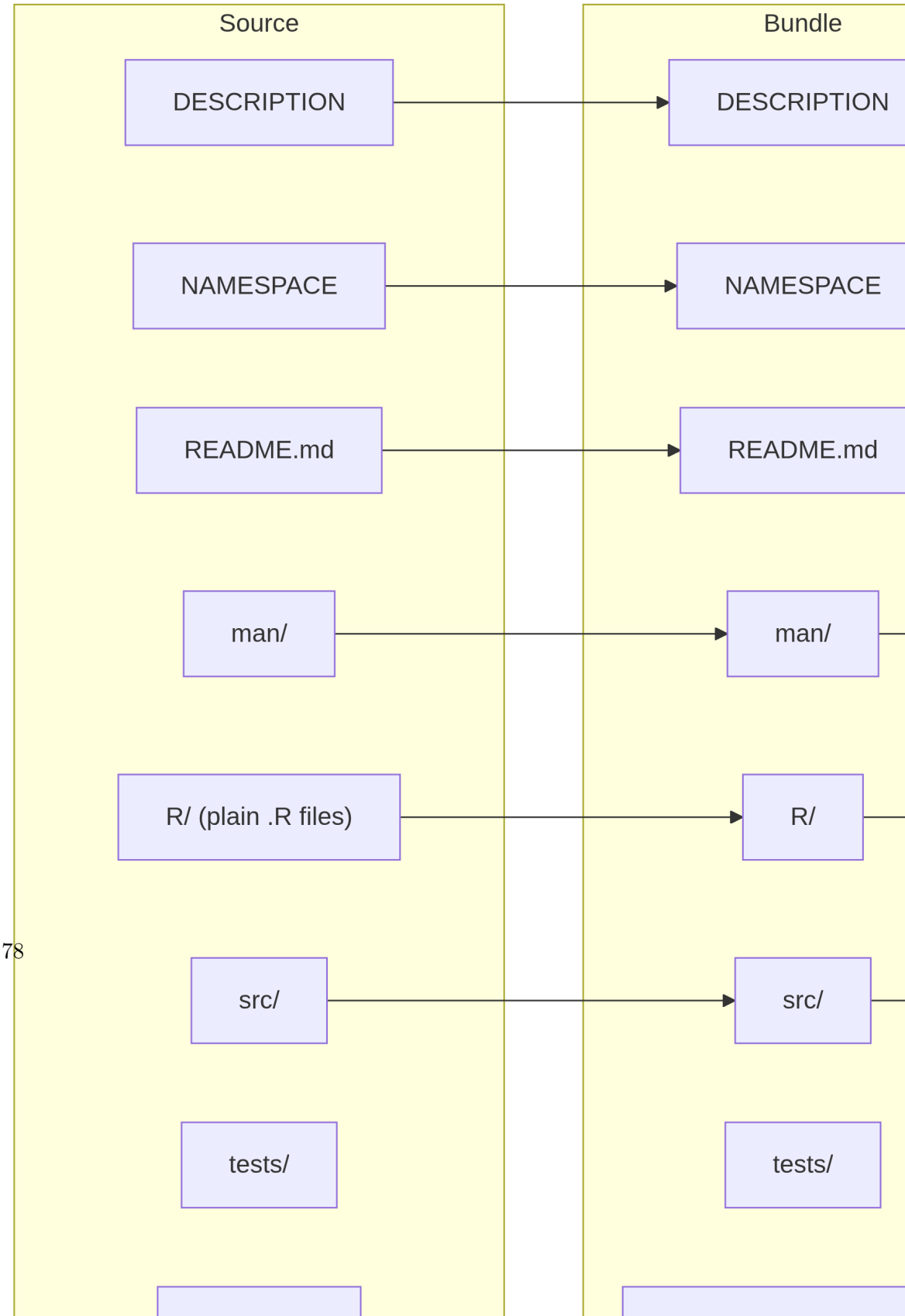
3.11 Lifecycle R development

The following diagrams summarise package construction, build, installation, and loading in an Excalidraw-like flow.

3.11.1 1) High-level package lifecycle



3.11.2 2) What changes between source, bundle, and binary



3.11.3 3) Command-driven workflow (construction to load)